



# DNA barcoding of meiofauna in a natural hydrocarbon seep

Bartłomiej Surmacz<sup>1</sup> · Daniele Camarda<sup>1</sup> · Antoni Żygadło<sup>3</sup> · Aleksander Gąsienica<sup>2</sup> · Daniel Stec<sup>1</sup>

Received: 2 January 2026 / Accepted: 2 July 2026 / Published online: 20 July 2026  
© The Author(s) 2026

## Abstract

Natural oil seeps remain poorly explored in terms of their biodiversity. Most studies on these habitats have focused on prokaryotic communities involved in hydrocarbon metabolism, whereas the microscopic invertebrates inhabiting these environments have received little attention, and existing observations rarely include species-level identifications. Here, we collected samples from a natural oil seep and characterized the meiofaunal community using DNA barcoding to support species identification. We detected representatives of four animal phyla known for their tolerance of extreme conditions: Annelida, Nematoda, Rotifera and Tardigrada. Across these groups, we identified at least seven putative species, including several that may represent previously unrecognized taxa. The meiofaunal assemblage comprised both widespread species and rarer taxa occasionally found in other habitats. Although hydrocarbon-polluted environments likely restrict many species, they also appear to provide ecological niches for extremophile animals and may serve as important reservoirs of specialized biodiversity. Nevertheless, the association between meiofaunal taxa and seep habitats requires further validation and investigations. Overall, meiofauna offer a promising model for eco-evolutionary studies of hydrocarbon-impacted environments and for applications in biomonitoring and bioremediation.

**Keywords** Petroleum · Seepage · Biodiversity · Nematoda · Tardigrada · Rotifera

## Introduction

Biodiversity is inherently connected with the abiotic environment, and geodiversity (defined as variation in geology, soils, and geomorphology) forms a fundamental basis for biological diversity (Tukiainen et al. 2023). While naturally geodiverse areas are typically associated with scenic, heterogeneous landscapes, other geodiverse habitats may appear visually unattractive, complicating efforts to justify their conservation. One such example is natural petroleum seepages. These environments are chronically polluted by hydrocarbons, which affect the physicochemical properties of the soil and influence biological communities either directly—through toxic effects such as disruption of cell membranes (Sikkema et al. 1995), or indirectly by altering microbial

activity and reshaping trophic interactions (Brázová et al. 2022). Despite the generally negative impact of oil seeps on many organisms, various taxa are able to persist in such polluted environments. Some extremophile taxa are particularly adapted to these conditions, and their distribution is linked to hydrocarbon-polluted habitats. The most thoroughly studied organisms associated with oil seeps are bacteria capable of metabolizing hydrocarbons. These prokaryotes have high potential for applications in bioremediation and have attracted considerable attention from researchers investigating their diversity and metabolic pathways (e.g., Heider et al. 1998; Guo et al. 2012). In contrast, the eukaryotes inhabiting these environments have received far less attention, and their communities remain poorly explored.

An interesting group of animals occurring in such habitats is meiofauna, defined as assemblages of aquatic or semiaquatic microinvertebrates smaller than 1 mm, encompassing groups such as nematodes, tardigrades, and rotifers (Remelli et al. 2020). Due to their small size, short life cycles, and large population sizes, meiofaunal species have high adaptive potential and are gaining attention in eco-evolutionary studies and bioindication (Martínez et al. 2025). While some research has documented microscopic invertebrates

✉ Bartłomiej Surmacz  
b.surmacz@isez.pan.krakow.pl

<sup>1</sup> Institute of Systematics and Evolution of Animals, Polish Academy of Sciences, Kraków, Poland

<sup>2</sup> Polish Geological Institute – National Research Institute, Carpathian Branch, Kraków, Poland

<sup>3</sup> Kraków, Poland

in seep sites, observations of meiofauna from oil seeps remain scarce and are typically limited to reporting taxa at the phylum level. Remelli et al. (2020) observed nematodes, tardigrades, and rotifers in terrestrial petroleum seeps but did not provide species-level identifications, constraining the ecological inferences that can be drawn about meiofauna in hydrocarbon-impacted environments. To address these limitations, we conducted the first species-level survey of meiofaunal communities inhabiting a natural oil seep, using DNA barcoding to enhance identification at lower taxonomic levels. Our aim was to investigate whether the species found in such habitats represent widespread generalist taxa or whether they include potential specialists adapted to hydrocarbon-impacted environments.

## Materials and methods

### Sampling site and geological setting

In the Polish Carpathians, the Grybów region is one of the most abundant areas in terms of bitumen occurrences (Świdziński 1948). Their presence is most commonly observed as impregnations within rock fractures and pore spaces, and in some cases also as natural petroleum seepages and gas exhalations (Gašienica and Bładusiak 2024). Crude oil in the Grybów area was extracted between 1900 and 1969 from the Grybów – Stara Wieś – Posadowa oil deposit (GSWP). The remnants of past exploitation include old pits and wells, typically located at sites of natural oil seepages (Świdziński 1948; Karnkowski and Konarski 1973). Even after oil production ceased, seepages remain active, and local residents still collect crude oil for their own purposes (e.g., for wood impregnation).

The geological structure of the Grybów area is associated with two tectonic units of the Polish Carpathians, i.e., the Magura and Grybów units (Paul 1993; Fig. 1). These units significantly differ in both the rock series that compose them and in their structural positions within the orogen. The Magura Unit is overthrust onto the Grybów Unit from the south and acts as a seal for migrating hydrocarbons. As a result of geological processes such as erosion and tectonism, formations of the Grybów Unit have been exposed within the Magura Unit in the Grybów region, forming the so-called Grybów tectonic window (Kozikowski 1953; Kozikowski 1956; Sikora 1960; Paul 1993). Within this tectonic window, the Oligocene Menilite and Krosno Beds are exposed (Dziadzio and Matyasik 2018; Fig. 1), although their classification (lithostratigraphy) has been subject to various interpretations in the past (Świdziński 1948; Kozikowski 1953, 1956; Sikora 1960; Paul 1993). Geochemical studies of the Menilite Beds in the Grybów tectonic window have shown that these shales are characterized by an elevated content of

organic matter and reached the oil window – temperatures sufficient for the generation of liquid hydrocarbons – in the geological past (Dziadzio and Matyasik 2018).

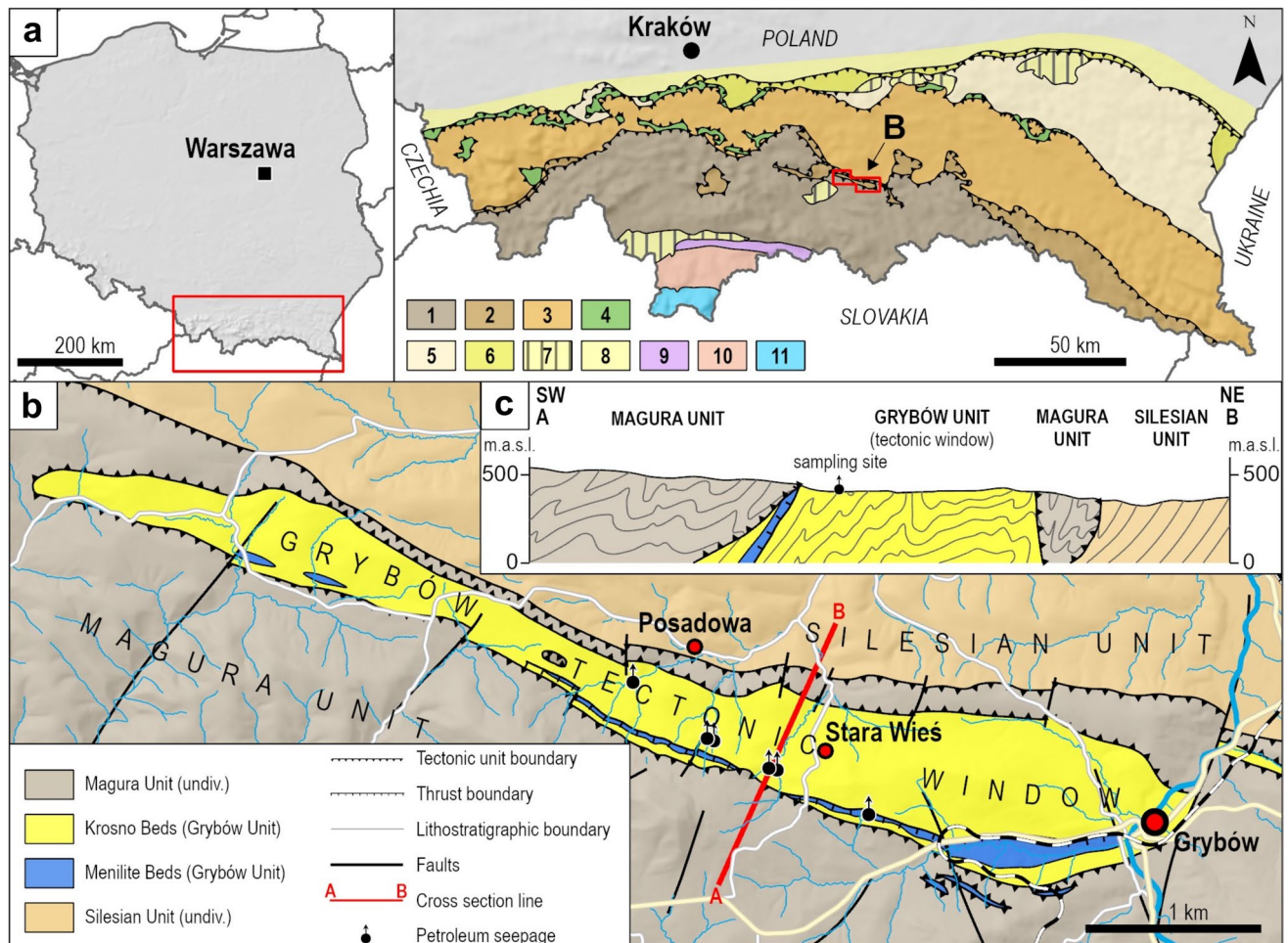
Within the Grybów tectonic window, one distinct thrust fault and a series of smaller, less traceable ones occur, the latter obscured by the lithological monotony of the Krosno Beds. These thrusts and associated fracture zones provide pathways for oil migrating from greater depths. The GSWP oil deposit occurs within thin-bedded sandstones and within fissures (Karnkowski and Konarski 1973). No recent studies have addressed the geochemical characteristics of oil from the GSWP deposit, and earlier documentation is fragmentary. According to Świdziński (1948), the GSWP deposit contains low-density (light) crude oil consisting primarily of benzene–naphthene fractions. The naturally seeping oil is reddish-green in color (Fig. 2).

### Sampling and sample processing

Sampling was conducted on 14 May 2025 at an oil seep site located in a stream in Stara Wieś (Carpathian Mountains, Poland), 49.633° N, 20.878° E (Fig. 3). Two samples (200 ml of substrate each) were collected: one directly from decaying organic matter submerged in hydrocarbon-contaminated water (Sample PL.072), and a second consisting of soil collected 0.5 m away from the first sample, also visibly contaminated by petroleum outflow (Sample PL.073). Both samples were collected into sterile plastic jars and transported to the laboratory at the Institute of Systematics and Evolution of Animals, Polish Academy of Sciences in Kraków, Poland (ISEA PAS). The samples were processed immediately to facilitate the identification of active meiofaunal organisms. The samples were sieved using a set of metal sieves (500, 250, and 36 µm meshes). The sediment retained on the finest sieve was then transferred to a Falcon tube. Portions of the sieved material were transferred to Petri dishes, diluted with distilled water, and examined under a stereomicroscope.

### Microscopy and imaging

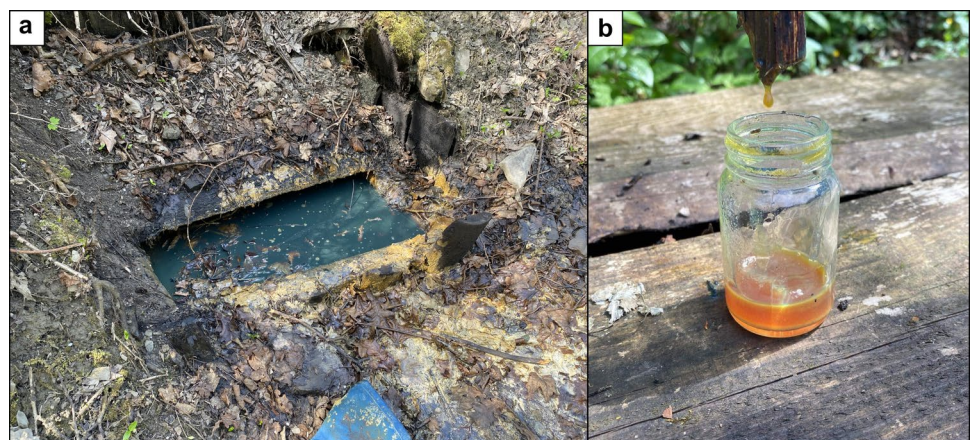
Several meiofaunal specimens representing different morphological groups were recognized and isolated under a stereomicroscope; then temporary water mounts and permanent slides using Hoyer's medium were prepared for observation with a light microscope. The animals mounted in temporary and permanent slides were examined and photographed under a ZEISS AX10 phase-contrast and differential interference contrast microscope equipped with a DLT-Cam PRO 20 MP digital camera. Figures were assembled with GIMP (GNU Image Manipulation Program; <https://gimp.org>).



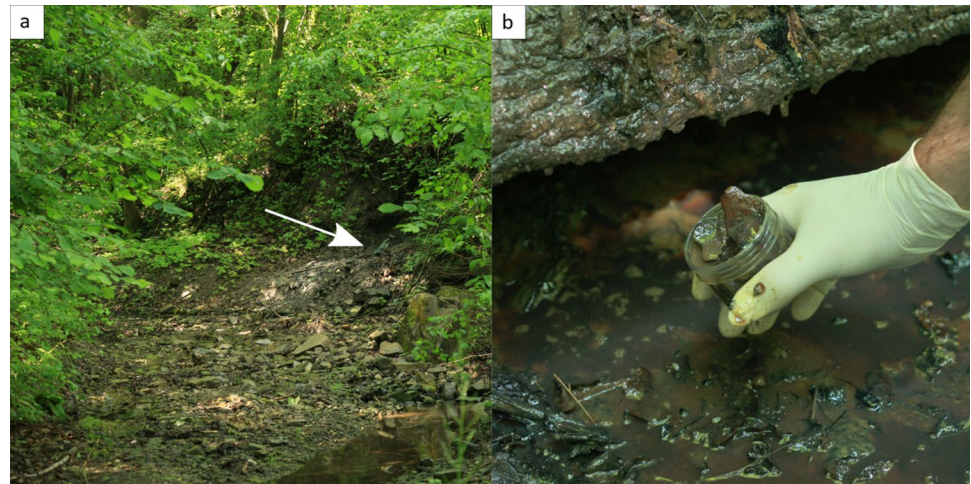
**Fig. 1** Location of the study area. **a** Geological map of the Polish Carpathians: 1 – Magura Unit; 2 – Dukła–Grybów and Foremagura Units; 3 – Silesian Unit; 4 – Subsilesian Unit; 5 – Skole Unit; 6 – Stebnik and Zgłobice Units; 7 – transgressive Miocene; 8 – Carpathian Foredeep; 9 – Pieniny Klippen Belt; 10 – Central Carpathian

Paleogene; 11 – Inner Carpathian Units (Żytko et al. 1989; modified). **b** Simplified geological map of the Grybów tectonic window (Świdziński 1948; Burtan et al. 1981; Paul 1993; modified). **c** Simplified geological cross-section (Świdziński 1948; modified)

**Fig. 2** Oil pits in Stara Wieś near Grybów, located within the Ropnik Stream basin. **a** Deep pit on the left bank filled with greenish crude oil; **b** sample of reddish crude oil collected from a pit on the right bank of the stream



**Fig. 3** Study area: **a** overview of sampling site; the arrow indicates the collection site; **b** collection of the substrate (decaying organic matter in oil-contaminated water)



## DNA barcoding

Animals representing distinctive morphospecies were then selected for DNA barcoding. DNA was extracted from individual animals following the Chelex® 100 resin (Bio-Rad) extraction method (Casquet et al. 2012), with modifications by Stec et al. (2020). The 658 base pair long fragment of the cytochrome c oxidase subunit I (COI) gene was amplified using the LCO1490-JJ and HCO2198-JJ primer set (Astrin and Stüben 2008). The amplification and sequencing were done according to the protocols described in Stec et al. (2020). Sequencing products were purified with the ExTerminator kit (A&A Biotechnology) and read with the ABI 3130xl sequencer. The sequences were processed in BioEdit 7.2.5 (Hall 1999) and submitted to GenBank (accession numbers PX369822–PX369853). Before submission, all obtained COI sequences were translated into protein sequences in AliView 1.28 (Larsson 2014) to check against pseudogenes. The obtained sequences were used as queries in an NCBI BLAST (Altschul et al. 1990) search to assess their similarity to sequences already present in GenBank. Similar sequences were considered as conspecific when having at least 97% sequence identity.

## Results

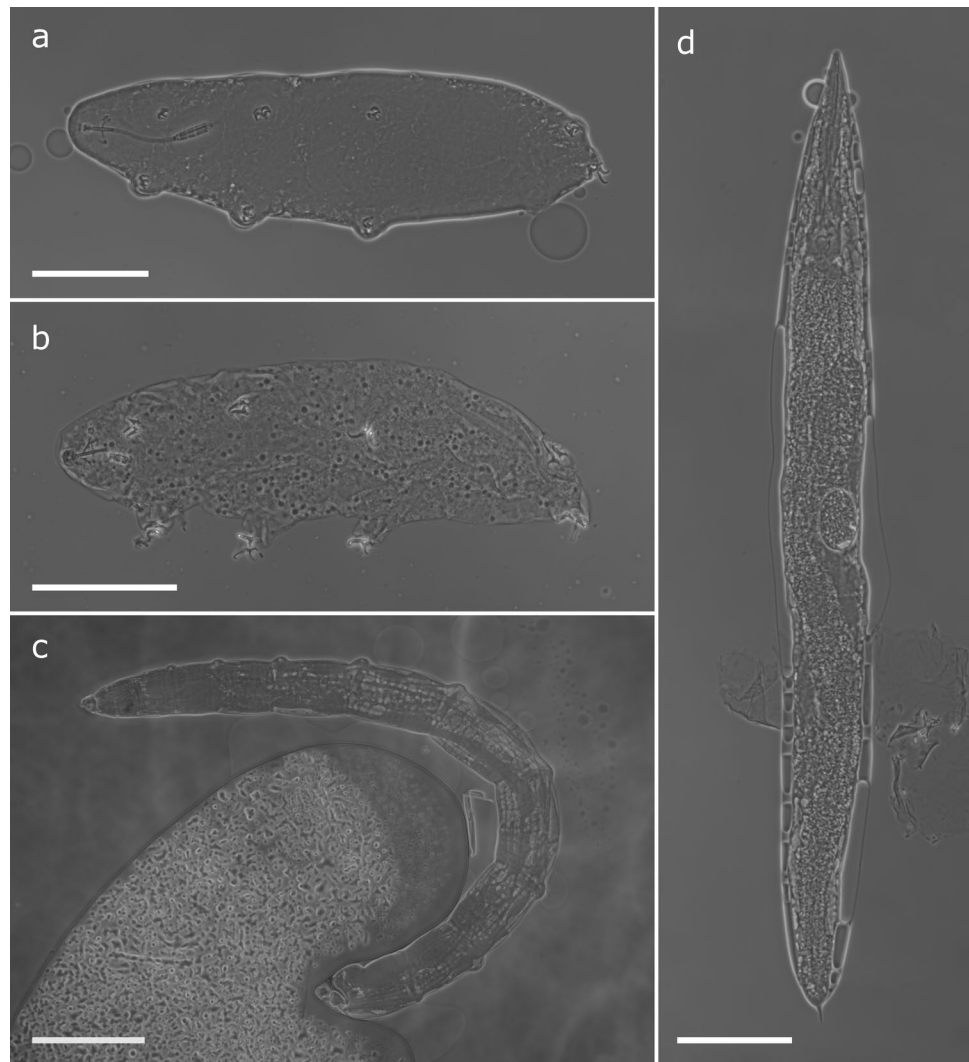
### Invertebrate community composition

Both investigated samples (PL.072 and PL.073) contained abundant populations of pot worms (Oligochaeta: Enchytraeidae) and nematodes, while rotifers and tardigrades were present but less abundant (Table 1). The microphotographs of representatives of all the animal phyla found in the samples are presented in the Fig. 4. We successfully amplified and sequenced COI barcodes for animals from each of these groups, generating a total of 32 sequences (Nematoda – 10, Oligochaeta – 5, Rotifera – 3, Tardigrada – 14, Table 2). All examined oligochaetes were classified as members of the *Enchytraeus buchholzi* Vejdovský, 1879 species complex, a result supported by DNA barcoding (>97% identity in BLAST searches). The dominant nematode genus (based on morphology) was *Phasmarhabditis*, with the closest COI match (92% identity) corresponding to *Phasmarhabditis bohémica* Nermut, Půža, Mekete & Mráček, 2017. In addition, we recorded *Seinura italiensis* Gu, Maria, Liu & Pedram, 2020. For rotifers, their COI sequences matched reference sequences with only 90–93%

**Table 1** Summary of the taxa abundance found in this study. Classification of estimated species abundance in a 200 ml sample: abundant (>500 individuals), moderate (10–500 individuals), rare (<10 individuals)

| Taxon                                   | Sample PL.072<br>organic matter sampled directly from the seepage, submerged in oil-polluted water | Sample PL.073<br>soil visibly contaminated by the seepage |
|-----------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Annelida: Enchytraeidae                 | Abundant                                                                                           | Abundant                                                  |
| Nematoda: Aphelenchoididae, Rhabditidae | Abundant                                                                                           | Abundant                                                  |
| Rotifera: Bdelloidea                    | Moderate                                                                                           | Moderate                                                  |
| Tardigrada: <i>Diphascon</i> spp.       | Moderate                                                                                           | Moderate                                                  |
| Tardigrada: <i>Hypsibius</i> spp.       | -                                                                                                  | Rare                                                      |
| Tardigrada: <i>Macrobiotus ripperi</i>  | -                                                                                                  | Rare                                                      |

**Fig. 4** Examples of species found in the study: **a** *Diphascon higginsii* (Tardigrada); **b** *Hypsibius* sp. (Tardigrada); **c** *Enchytraeus buchholzi* agg. (Anellida); **d** *Phasmarhabditis* sp. (Nematoda). Scale bars: 100  $\mu$ m (a–c); 250  $\mu$ m (d)



**Table 2** GenBank accession numbers for COI sequences of specimens barcoded in this study, along with the best hits from the BLAST sequence similarity queries

| Phylum     | Taxonomic classification             | GenBank accession numbers                                  | BLAST best hits (sequence % identity)                                                                                        |
|------------|--------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Annelida   | <i>Enchytraeus buchholzi</i> complex | PX369822-PX369826                                          | <i>Enchytraeus buchholzi</i> complex<br>PP180581, PP180537 (>97%)                                                            |
| Nematoda   | <i>Seinura italiensis</i>            | PX369829-PX369836                                          | <i>Seinura italiensis</i> LC596446 (>97%)                                                                                    |
|            | <i>Phasmarhabditis</i> sp.           | PX369827-PX369828                                          | <i>Phasmarhabditis</i> sp. KX576459 (92%)                                                                                    |
| Rotifera   | Bdelloidea (unidentified)            | PX369837-PX369839                                          | Unidentified Bdelloidea EF650542 (93%)<br><i>Harbotrocha constricta</i> JX184002 (91%)<br><i>Adineta vaga</i> EF173224 (90%) |
| Tardigrada | <i>Diphascon higginsii</i>           | PX369843, PX369845, PX369847-PX369849                      | <i>Diphascon higginsii</i> KU513417 (>97%)                                                                                   |
|            | <i>Diphascon</i> sp.                 | PX369840- PX369842, PX369844, PX369847, PX369850, PX369846 | <i>Diphascon higginsii</i> KU513417 (87%)                                                                                    |
|            | <i>Hypsibius</i> sp. 1               | PX369851                                                   | <i>Hypsibius</i> sp. PQ140631 (99%)<br><i>Hypsibius</i> sp. PX701228 (99%)                                                   |
|            | <i>Hypsibius</i> sp. 2               | PX369852                                                   | <i>Hypsibius</i> sp. PQ140633 (87%)                                                                                          |
|            | <i>Macrobiotus ripperi</i>           | PX369853                                                   | <i>Macrobiotus ripperi</i> MN888312 (99%)                                                                                    |

identity, preventing reliable identification at the species or genus level. Tardigrades were represented only by the class Eutardigrada, including specimens belonging to the genus *Diphascon* Plate, 1888 (the most abundant), and a few individuals of the genera *Hypsibius* Ehrenberg, 1848 and *Macrobiotus* C.A.S. Schultze, 1834. The DNA barcoding of the tardigrade specimens revealed the presence of *Diphascon higginsi* Binda, 1971 (>97% identity; a result also supported by morphological analysis; Online resource) and one additional, unclassified *Diphascon* species. Among *Hypsibius*, we identified two putative species, including one whose COI sequence matched an existing reference sequence (Table 2).

## Discussion

### Invertebrate communities of oil seep site

Our study presents a superficial insight into microinvertebrate communities inhabiting a site naturally impacted by hydrocarbon outflow. While focusing on meiofauna, we also observed slightly larger animals that do not strictly fall within the definition of meiofauna, namely pot worms (Annelida: Enchytraeidae). These animals were found at high abundances in both samples and were classified as members of the *Enchytraeus buchholzi* species complex. Enchytraeids have been frequently used in soil bioindication (e.g., Pelosi and Römbke 2016), including assessments of oil contamination. The effects of petroleum contamination on soils have been studied in the related species *Enchytraeus albidus* Henle, 1837, which showed reduced survival and reproduction at elevated hydrocarbon concentrations (Kovaleva et al. 2017). However, another species, *Enchytraeus doerjesi* Westheide & Graefe, 1992, was found to be less sensitive than other soil invertebrates when exposed to oil refinery sludge (Reinecke et al. 2016). Our finding of an abundant population of the *E. buchholzi* complex in an oil-impacted site

suggests a comparatively high tolerance to hydrocarbon pollution, highlighting the need for accurate species identification and a better understanding of species-specific ecological tolerance when using non-standardized taxa as bioindicators in soil ecotoxicology (Alves et al. 2022). The dominant meiofaunal group recorded in our study was Nematoda, whereas rotifers and tardigrades were less abundant (Table 1). Interestingly, a similar pattern of meiofaunal composition was observed in Italian surface hydrocarbon seeps by Remelli et al. (2020). Nematodes are known to occur in petroleum-contaminated environments, and together with bacteria, they may contribute to bioremediation processes (Beyrem et al. 2010; Chan 2011). Tardigrades and rotifers formed lower population densities, suggesting that seep conditions may be less favorable for these groups.

### Species-level identifications

The main contribution of our study to the ecology of meiofauna in oil seep sites is the provision of species-level identifications obtained through DNA barcoding. The resulting patterns were unexpected, as we recorded species known from very distinct and geographically separated habitats (Table 3), as well as several species for which no reference barcodes are currently available. The taxa recovered in our study include several examples of species previously reported from other environments. For instance, the nematode *Seinura italiensis* was originally isolated from soil imported from Italy to China (Gu et al. 2020) and was later found in a tree hollow in Japan (Kanzaki et al. 2021). Another example is the tardigrade *Diphascon higginsi*, representing the most abundant tardigrade species in our samples, which was previously recorded from rotting leaf litter in a pond in a botanic garden approximately 100 km from our sampling site (Kosztyła et al. 2016). *Diphascon higginsi* was originally described from Morocco (Binda 1971), yet the description lacks molecular data, making it unclear

**Table 3** Summary of the distribution of taxa identified by DNA barcodes in this study. The biogeographic records are considered only when supported by molecular data

| Species                                | Biogeography                                                                                                                           |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Annelida: <i>Enchytraeus buchholzi</i> | Widespread, primarily found in the temperate Palearctic (Zhao et al. 2022)                                                             |
| Nematoda: <i>Seinura italiensis</i>    | Potentially widespread, recorded from soil imported from Italy to China and from Japan (Gu et al. 2020, Kanzaki et al. 2021)           |
| Nematoda: <i>Phasmarhabditis</i> sp.   | No other genetically confirmed records                                                                                                 |
| Rotifera: Bdelloidea                   | No other genetically confirmed records                                                                                                 |
| Tardigrada: <i>Diphascon higginsi</i>  | Recorded only from Poland so far (<100 km from the sampling site) (Kosztyła et al. 2016)                                               |
| Tardigrada: <i>Diphascon</i> sp. 2     | No other genetically confirmed records                                                                                                 |
| Tardigrada: <i>Hypsibius</i> sp. 1     | Recorded from Poland (<100 km from the sampling site) (Surmacz et al. 2026) and from an alpine peatland in Italy (Camarda et al. 2026) |
| Tardigrada: <i>Hypsibius</i> sp. 2     | No other genetically confirmed records                                                                                                 |
| Tardigrada: <i>Macrobiotus ripperi</i> | Potentially common in Europe, recorded from Poland and Finland (Stec et al. 2021; Surmacz et al. 2026)                                 |

whether the original material is conspecific with subsequent records or instead represents part of a species complex of morphologically similar taxa. Additional tardigrade species recorded elsewhere include *Hypsibius* sp. 1, previously found in a moss sample from an urban environment also ~100 km from our site (Surmacz et al. 2026) and in an alpine peatland in Italy (Camarda et al. 2026), as well as *Macrobiotus ripperi*, recorded from several localities in Poland and Finland (Stec et al. 2021; Surmacz et al. 2026).

Apart from the species recorded worldwide, potentially with broad ecological niches and geographic ranges, our analysis revealed the presence of previously unbarcoded species that may represent taxa new to science, that may be particularly adapted to hydrocarbon-polluted environments. Such examples include nematodes, tardigrades, and rotifers for which no COI reference barcodes are available in public databases (Table 2). However, for nematodes the COI reference availability is limited, as most molecular studies on nematodes target rRNA genes. Importantly, the absence of reference sequences does not constitute evidence of taxonomic novelty, and conclusions regarding this possibility should be evaluated in detail using an integrative taxonomic approach. Nevertheless, the reference DNA sequences provided here for the animals we encountered in our study will facilitate future comparisons of meiofaunal assemblages across other seep habitats, helping to assess the strength of species–microhabitat associations and to determine whether seep environments may select for particularly well-adapted taxa. Overall, our study not only generated novel occurrence data of meiofaunal species from a unique habitat, but also provided reference sequences for new taxa, contributing to the understanding of meiofauna spatial distributions.

## Conclusions

Our research addresses a key knowledge gap by providing species-level identifications of meiofauna inhabiting terrestrial hydrocarbon seeps, supported by newly generated DNA barcodes. The species recorded from the seep site include unexpected biogeographical occurrences of taxa previously known from distant localities, as well as putative undescribed species that had not been barcoded prior to this study. In many cases, species identifications were enabled or strengthened by COI barcoding, and even when definitive identifications were not possible, the barcode sequences themselves represent a valuable contribution to reference databases, facilitating future taxonomic and ecological work. We show that meiofauna represent a promising target for future investigations into eco-evolutionary processes of hydrocarbon-impacted environments and may also have potential applications in biomonitoring or bioremediation. Even though hydrocarbon seeps may reduce local diversity through their toxic effects, they can simultaneously

support specialized biodiversity by providing natural niches for extremophiles.

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1007/s11756-026-02261-x>.

**Acknowledgments** The study was supported by the Institute of Systematics and Evolution of Animals, Polish Academy of Sciences. BS is supported by the Foundation for Polish Science (FNP) START scholarship awarded in 2025.

**Author contributions** All authors contributed to the study conception and design. Sampling was done by Bartłomiej Surmacz, Antoni Żygadło, and Aleksander Gąsienica. The samples were investigated by Daniele Camarda, Daniel Stec, and Bartłomiej Surmacz. The DNA sequencing and imaging were done by Daniele Camarda and Daniel Stec. The first draft of the manuscript was written by Bartłomiej Surmacz, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

**Funding** The study was supported by the Institute of Systematics and Evolution of Animals, Polish Academy of Sciences. BS is supported by the Foundation for Polish Science (FNP) START scholarship awarded in 2025.

**Data availability** All the sequences generated in the study are deposited in NCBI Genbank (accession numbers PX369822–PX369853).

## Declarations

**Competing interests** The authors declare no competing interests.

**Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

## References

- Altschul SF, Gish W, Miller W, Myers EW, Lipman DJ (1990) Basic local alignment search tool. *J Mol Biol* 215(3):403–410. [https://doi.org/10.1016/S0022-2836\(05\)80360-2](https://doi.org/10.1016/S0022-2836(05)80360-2)
- Alves PRL, Bandeira FO, Hennig TB (2022) Ecological role of earthworms as bioindicators of soil health. In: Vig AP, Singh J, Suthar SS (eds) *Earthworm engineering and applications*. Nova Science Publishers, New York, pp 55–81. <https://doi.org/10.52305/VLNM4957>
- Astrin JJ, Stüben PE (2008) Phylogeny in cryptic weevils: molecules, morphology and new genera of western Palaearctic Cryptorhynchinae (Coleoptera: Curculionidae). *Invertebr Syst* 22(5):503–522. <https://doi.org/10.1071/IS07057>
- Beyrem H, Louati H, Essid N, Aïssa P, Mahmoudi E (2010) Effects of two lubricant oils on marine nematode assemblages in a laboratory microcosm experiment. *Mar Environ Res* 69(4):248–253. <https://doi.org/10.1016/j.marenvres.2009.10.018>

- Binda M (1971) Su alcuni tardigradi muscicoli del Nord-Africa. Bollettino delle sedute dell'Accademia Gioenia di Scienze Naturali, Catania 10:759–765
- Brázová T, Kováčik P, Matoušková M, Oros M (2022) Nematodes as soil stress indicators for polycyclic aromatic hydrocarbons: a review. *Helminthologia* 59(2):117–126. <https://doi.org/10.2478/helm-2022-0014>
- Burtan J, Golonka J, Oszczytko N, Paul Z, Ślaczka A (1981) Mapa Geologiczna Polski, skala 1: 200 000, ark. Nowy Sącz. Wydawnictwa Geologiczne
- Camarda D, Fontaneto D, Bajorek D, Lisi O, Stec D (2026) Taxonomic survey and DNA barcode library of tardigrades from alpine fens in northwestern Italy. *J Limnol* 85:2262. <https://doi.org/10.4081/jlimnol.2026.2262>
- Casquet J, Thebaud C, Gillespie RG (2012) Chelex without boiling, a rapid and easy technique to obtain stable amplifiable DNA from small amounts of ethanol-stored spiders. *Mol Ecol Resour* 12(1):136–141. <https://doi.org/10.1111/j.1755-0998.2011.03073.x>
- Chan H (2011) Biodegradation of petroleum oil achieved by bacteria and nematodes in contaminated water. *Sep Purif Technol* 80(3):459–466. <https://doi.org/10.1016/j.seppur.2011.05.028>
- Dziadzio PS, Matyasik I (2018) Środowisko sedimentacji i korelacja geochemiczna dolnooligocenów utworów z obrębu jednostek dukielskiej i grybowskiej. *Nafta-Gaz*. <https://doi.org/10.18668/ng.2018.06.02>
- Gąsienica A, Bładusiak P (2024) Karta wycieków węglowodorów – Grybów. Archiwum PIG-PIB, Warszawa, Poland
- Gu J, Maria M, Liu L, Pedram M (2020) Description of *Seinura italensis* n. sp. (Tylenchomorpha: Aphelenchoididae) found in the medium soil imported from Italy. *J Nematol* 52:e2020-18. <https://doi.org/10.21307/jofnem-2020-018>
- Guo H, Yao J, Cai M, Qian Y, Guo Y, Richnow HH, Blake RE, Doni S, Ceccanti B (2012) Effects of petroleum contamination on soil microbial numbers, metabolic activity and urease activity. *Chemosphere* 87(11):1273–1280. <https://doi.org/10.1016/j.chemosphere.2012.01.034>
- Hall TA (1999) BioEdit: a user-friendly biological sequence alignment editor and analysis program for Windows 95/98/NT. *Nucleic Acids Symp Ser* 41:95–98
- Heider J, Spormann AM, Beller HR, Widdel F (1998) Anaerobic bacterial metabolism of hydrocarbons. *FEMS Microbiol Rev* 22(5):459–473. <https://doi.org/10.1111/j.1574-6976.1998.tb00381.x>
- Kanzaki N, Ekino T, Hamaguchi K, Takeuchi-Kaneko Y (2021) Three *Seinura* species from Japan with a description of *S. shigaensis* n. sp. (Tylenchomorpha: Aphelenchoididae). *PLoS One* 16(1):e0244653. <https://doi.org/10.1371/journal.pone.0244653>
- Karnkowski P, Konarski E (1973) Katalog złóż ropy naftowej i gazu ziemnego w Polsce. Karpaty. Wydawnictwo Geologiczne, Warszawa
- Kosztyla P, Stec D, Morek W, Gąsiorek P, Zawierucha K, Michno K, Ufir K, Małek D, Hlebowicz K, Laska A, Dudziak M, Frohme M, Prokop ZM, Kaczmarek Ł, Michalczyk Ł (2016) Experimental taxonomy confirms the environmental stability of morphometric traits in a taxonomically challenging group of microinvertebrates. *Zool J Linn Soc* 178(4):765–775. <https://doi.org/10.1111/zoj.12409>
- Kovaleva E, Yakovlev A, Nikolaenko M, Makarov A, Makarov A (2017) Ecological evaluation of oil-contaminated soils (Sakhalin) using enchytraeidae. *Eurasian Soil Sci* 50(3):350–358. <https://doi.org/10.1134/S1064229317030073>
- Kozikowski H (1953) Budowa geologiczna okolic Klęczan-Pisarzowej. Wydawnictwo Geologiczne, Warszawa
- Kozikowski H (1956) Jednostka Ropy-Pisarzowej, nowa jednostka tektoniczna polskich Karpat fliszowych. *Biul Inst Geol* 110:93–119
- Larsson A (2014) Aliview: a fast and lightweight alignment viewer and editor for large datasets. *Bioinformatics* 30(22):3276–3278. <https://doi.org/10.1093/bioinformatics/btu531>
- Martínez A, Bonaglia S, Di Domenico M, Fonseca G, Ingels J, Jörger KM et al (2025) Fundamental questions in meiofauna research highlight how small but ubiquitous animals can improve our understanding of nature. *Commun Biol* 8(1):449. <https://doi.org/10.1038/s42003-025-07888-1>
- Paul Z (1993) Szczegółowa mapa geologiczna Polski – arkusz Grybów (wraz z objaśnieniami). Państw. Inst. Geol., Warszawa, Poland
- Pelosi C, Römbke J (2016) Are *Enchytraeidae* (Oligochaeta, Annelida) good indicators of agricultural management practices? *Soil Biol Biochem* 100:255–263. <https://doi.org/10.1016/j.soilbio.2016.06.030>
- Reinecke AJ, van Wyk M, Reinecke SA (2016) Toxicity to earthworms and chemical composition of oil refinery sludge destined for land-farming. *Soil Sediment Contam* 25(8):868–881. <https://doi.org/10.1080/15320383.2016.1218823>
- Remelli S, Rizzo P, Celico F, Menta C (2020) Natural surface hydrocarbons and soil faunal biodiversity: a bioremediation perspective. *Water* 12(9):2358. <https://doi.org/10.3390/w12092358>
- Sikkema J, de Bont JA, Poolman B (1995) Mechanisms of membrane toxicity of hydrocarbons. *Microbiol Rev* 59(2):201–222. <https://doi.org/10.1128/mr.59.2.201-222.1995>
- Sikora W (1960) O stratygrafii serii okiennej w Ropie koło Gorlic. *Geol Q* 4(1):153–172
- Stec D, Kristensen RM, Michalczyk Ł (2020) An integrative description of *Minibiotus ioculator* sp. nov. from the Republic of South Africa with notes on *Minibiotus pentannulatus* Londoño et al., 2017 (Tardigrada: Macrobiotidae). *Zool Anz* 286:117–134. <https://doi.org/10.1016/j.jcz.2020.03.007>
- Stec D, Vecchi M, Dudziak M, Bartels PJ, Calhim S, Michalczyk Ł (2021) Integrative taxonomy resolves species identities within the *Macrobiotus pallarii* complex (Eutardigrada: Macrobiotidae). *Zool Lett* 7(1):9. <https://doi.org/10.1186/s40851-021-00176-w>
- Surmacz B, Vecchi M, Fontaneto D, Budzik K, Godziek J, Matsko Y, Stec D (2026) COI metabarcoding with a curated reference database and optimized protocol provides a reliable species-level diversity assessment of tardigrades. *Integr Zool* 21(2):275–290. <https://doi.org/10.1111/1749-4877.12972>
- Świdziński H (1948) Strefa naftowa Grybów-Stara Wieś-Posadowa. Arch. PIG-PIB Oddz. Karpacki
- Tukiaainen H, Toivanen M, Maliniemi T (2023) Geodiversity and biodiversity: Geological Society, London, Special Publications 530:31–47. <https://doi.org/10.1144/sp530-2022-107>
- Zhao L, Xie X, Chen D, Ma G, Sun Y (2022) Microscopic observations on form and structure of the worm *Enchytraeus buchholzi* (Clitellata: Enchytraeidae). *BMC Zool* 7(1):31. <https://doi.org/10.1186/s40850-022-00123-y>
- Żytko K, Zajac R, Gucik S, Rylko W, Oszczytko N, Garlicka I et al (1989) Map of the tectonic elements of the Western Outer Carpathians and their foreland. In: Poprawa D, Nemčok J (eds) *Geological Atlas of the Western Outer Carpathians and their Foreland*. Państwowy Instytut Geologiczny Warszawa/GUDŠ Bratislava/Ug Praha